

Unit 5.4 The Fundamental Theorem of Calculus p. 391 - 393

Note Title

2014/05/05

FTC 2

The Fundamental Theorem of Calculus, Part 2 If f is continuous on $[a, b]$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

where F is any antiderivative of f , that is, a function such that $F' = f$.

Proof: Divide $[a, b]$ into n subintervals of equal length $\Delta x = \frac{b-a}{n}$.

Let $x_0 = a < x_1 < \dots < x_{n-1} < x_n = b$ be the endpoints of the interval. (A)

Let F be an antiderivative of f so that $F' = f$ (B)

Therefore:

- F is differentiable on $[a, b]$ and
 - F is continuous on $[a, b]$
- $[f \text{ diff} \Rightarrow f \text{ continuous}]$

Use the MVT on each subinterval $[x_{i-1}, x_i]$

1. F is continuous on $[x_{i-1}, x_i]$
2. F is differentiable on (x_{i-1}, x_i)

therefore by the MVT there exists a number $c_i \in (x_{i-1}, x_i)$ such that

$$\frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} = F'(c_i) = f(c_i) \quad (C) \quad \text{from (B).}$$

Let these c_i 's be the sample point in each interval.

As f is continuous on $[a, b]$, f is integrable on $[a, b]$ - Theorem p 373

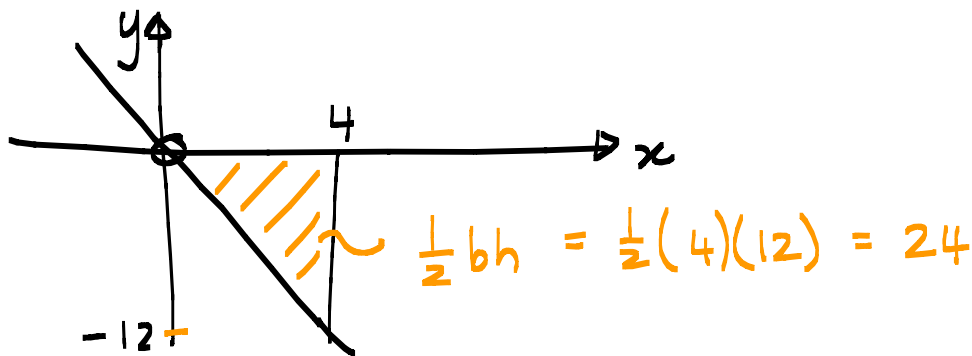
Using definition p 374:

$$\begin{aligned} \int_a^b f(x) dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(c_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{F(x_i) - F(x_{i-1})}{x_i - x_{i-1}} (x_i - x_{i-1}) \quad \text{From C} \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n [F(x_i) - F(x_{i-1})] \\ &= \lim_{n \rightarrow \infty} [\cancel{F(x_1)} - F(x_0) + F(x_2) - \cancel{F(x_1)} + \dots \\ &\quad + F(x_n) - F(x_{n-1})] \\ &= \lim_{n \rightarrow \infty} [F(x_n) - F(x_0)] \\ &= \lim_{n \rightarrow \infty} [F(b) - F(a)] \\ &= F(b) - F(a) \end{aligned}$$

Examples:

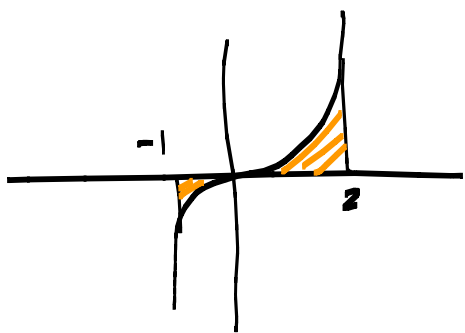
$$\begin{aligned}
 1. \text{ Find } \int_0^4 -3x \, dx &= -\frac{3}{2} x^2 \Big|_0^4 \\
 &= -\frac{3}{2} (4^2 - 0^2) = -24
 \end{aligned}$$

In terms of the area:



2. Calculate the area of the region between the graph of $f(x) = x^3$, the x -axis and $x = -1$ and $x = 2$.

We use the definite integral to calculate the area.



$$\begin{aligned}
 \int_{-1}^0 x^3 \, dx &= \frac{1}{4} x^4 \Big|_{-1}^0 \\
 &= 0 - \frac{1}{4} (-1)^4 = -\frac{1}{4} \\
 \int_0^2 x^3 \, dx &= \frac{1}{4} x^4 \Big|_0^2 \\
 &= \frac{16}{4} - 0 = 4
 \end{aligned}$$

$$\text{The area} = 4 + \frac{1}{4} = \frac{17}{4}$$

3. Calculate

$$1. \int_{-1}^2 e^{5x} \, dx = \frac{1}{5} e^{5x} \Big|_{-1}^2 = \frac{1}{5} (e^{10} - e^{-5})$$

$$2. \int_{-2}^1 x^2 e^{x^3} dx = \frac{1}{3} e^{x^3} \Big|_{-2}^1 = \frac{1}{3} (e - e^{-8})$$

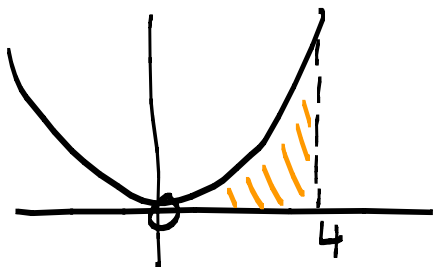
$$3. \int_0^{\pi} \cos 3x dx = \frac{1}{3} \sin 3x \Big|_0^{\pi} = 0$$

$$4. \int_0^1 2^t \cos 2^t dt = \frac{\sin 2^t}{\ln 2} \Big|_0^1 = \frac{\sin 2}{\ln 2} - \frac{\sin 1}{\ln 2}$$

$$5. \int_{-1}^1 x \sqrt{1+x^2} dx = \frac{1}{2} \cdot \frac{2}{3} (1+x^2)^{3/2} \Big|_{-1}^1 = 0$$

$$6. \int_{-2}^{-1} \sqrt{1-x} dx = -\frac{2}{3} (1-x)^{3/2} \Big|_{-2}^{-1} \\ = -\frac{2}{3} [2^{3/2} - 3^{3/2}]$$

7. Find the area under the parabola $y = x^2$ between $x = 0$ and $x = 4$.



$$A = \int_0^4 x^2 dx \\ = \frac{1}{3} x^3 \Big|_0^4 = 21.33$$

$$8. \text{ Evaluate } \int_1^2 \frac{1}{x} dx = \ln|x| \Big|_1^2 \\ = \ln 2 - \ln 1 = \ln 2$$

9. What's wrong?

$$\int_{-1}^3 \frac{1}{x^2} dx = \int_{-1}^3 x^{-2} dx = -x^{-1} \Big|_{-1}^3 \\ = -\frac{1}{3} - 1 = -\frac{4}{3}$$

and yet



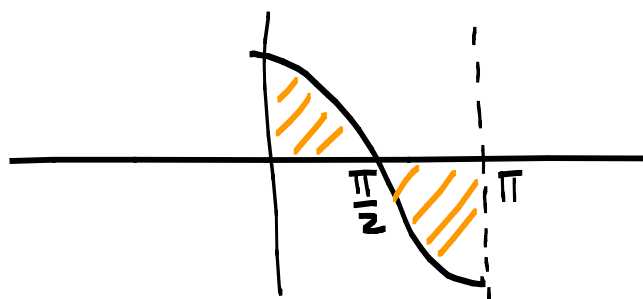
If you can determine the area it should be positive ???

What wrong? $\frac{1}{x^2}$ is not continuous on $[-1, 3]$, which is required by FTC 2 so we may not use the theorem.

10. Find the area between the curve and the x -axis

① $y = \cos x$ where $0 \leq x \leq \frac{\pi}{2}$ and

② $y = \cos x$ where $0 \leq x \leq \pi$



$$\begin{aligned} \text{① } & \int_0^{\frac{\pi}{2}} \cos x \, dx \\ &= \sin x \Big|_0^{\frac{\pi}{2}} = 1 \end{aligned}$$

$$\text{② } \int_0^{\pi} \cos x \, dx = 0 ?? \quad (\text{Check!!})$$

$$\begin{aligned}
 \text{Area} &= \int_0^{\frac{\pi}{2}} \cos x \, dx - \int_{\frac{\pi}{2}}^{\pi} \cos x \, dx \\
 &= \sin x \Big|_0^{\frac{\pi}{2}} - \sin x \Big|_{\frac{\pi}{2}}^{\pi} \\
 &= 1 - [0 - 1] = 2
 \end{aligned}$$